

Ejercicios de regla de la cadena

Instrucción: Derive cada una de las siguientes funciones.

Ejercicios

1. $f(x) = (3x + 2)^5$
2. $g(x) = (x^2 - 4x + 1)^3$
3. $h(x) = \sqrt{5x^2 + 1}$
4. $k(x) = \sin(x^3 + 2x)$
5. $m(x) = e^{\cos(4x)}$
6. $p(x) = \ln(\sqrt{x^2 + 1}) \cdot \tan(3x)$
7. $q(x) = \frac{\sin^4(3x)}{\sqrt{x^2 + 4}}$
8. $r(x) = \arctan(\sqrt{x^2 + 1}) \cdot e^{\cos^2(2x)}$
9. $s(x) = \arctan(e^{-3x}) \cdot \sin^3(\ln(x^2 + 1))$
10. $t(x) = \frac{\arctan(\sqrt{x}) \cdot \cos^3(e^{2x})}{\ln(x^2 + \sin^2 x + 1)}$

Respuestas

$$1. f'(x) = 15(3x + 2)^4$$

$$2. g'(x) = 6(x - 2)(x^2 - 4x + 1)^2$$

$$3. h'(x) = \frac{5x}{\sqrt{5x^2 + 1}}$$

$$4. k'(x) = (3x^2 + 2) \cos(x^3 + 2x)$$

$$5. m'(x) = -4 \sin(4x) e^{\cos(4x)}$$

$$6. p'(x) = \frac{x \tan(3x)}{x^2 + 1} + \frac{3}{2} \ln(x^2 + 1) \sec^2(3x)$$

$$7. q'(x) = \frac{\sin^3(3x) [12(x^2 + 4) \cos(3x) - x \sin(3x)]}{(x^2 + 4)^{3/2}}$$

$$8. r'(x) = e^{\cos^2(2x)} \left[\frac{x}{(x^2 + 2)\sqrt{x^2 + 1}} - 2 \sin(4x) \arctan(\sqrt{x^2 + 1}) \right]$$

$$9. s'(x) = \sin^2(\ln(x^2 + 1)) \left[\frac{-3e^{-3x} \sin(\ln(x^2 + 1))}{1 + e^{-6x}} + \frac{6x \cos(\ln(x^2 + 1)) \arctan(e^{-3x})}{x^2 + 1} \right]$$

10.

$$t'(x) = \frac{\cos^2(e^{2x}) \left[\frac{\cos(e^{2x})}{2\sqrt{x}(1+x)} - 6e^{2x} \sin(e^{2x}) \arctan(\sqrt{x}) \right] \ln(x^2 + \sin^2 x + 1)}{\ln^2(x^2 + \sin^2 x + 1)} - \frac{\arctan(\sqrt{x}) \cos^3(e^{2x}) \frac{2x + \sin(2x)}{x^2 + \sin^2 x + 1}}{\ln^2(x^2 + \sin^2 x + 1)}$$